

Inspection and Quality Control Robots

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Abstract— Inspecting has always been important to the reliability of a finished product. As manufacturing has scaled up, the methods used to inspect have evolved from handheld optical tools to sensor driven automated systems capable of detecting a micro dent in an automobile body or a damaged kernel in a high-speed grain sorting line. The demand for faster, more consistent, more reliable, and more adaptable inspection has created the need for robotics systems that can handle varied geometries, materials, and defect types without requiring manual reconfiguration between tasks. This project's aim is to design and evaluate an inspection and quality control robot framework capable of performing surface defect detection and to inspect parts and check their dimensions autonomously across a defined workspace.

The significance of such a system is straightforward. Manufacturing environments that depend on human inspection have always had a ceiling. Human inspectors face unavoidable limits, they get tired, fatigue, miss things, and the physical constraints of manual access to complex geometries all contributes to the inspection gaps that translates into field failures, recalls, rework cost, and compliance risk. A robotic system that delivers repeatable, sensor driven evaluation removes these variables and produces a detailed record for every part it assesses, which is increasingly a necessity requirement in regulated industries such as aerospace, automotive, and food processing.

Prior work has approached this problem from several directions, each solving part of it. Fixed-mount vision systems work best on a uniform surface under controlled lighting but breaks down when introduced to different defects and geometrical structures. Following this, mobile platforms were introduced with extended coverage to large structures but were limited by positional uncertainty that affects measurements' accuracy. Learning based methods improve adaptability but depend on a database of labeled defects that are costly to assembled in low volume settings and time consuming. These projects draw from combining adaptive sensing with kinematic control on a single platform rather than treating the detection and measurement as separate systems.

The expected outcome is a framework that operates autonomously, captures surface data from multiple angles, and labels each defect with accuracy and cycle time suitable for the production line. Every evaluated part produces a traceable inspection record, to offer better results towards consistent, scalable quality assurances, especially in an environment where existing single purpose inspection tools fall short.

Keywords—*Robotic inspection, Quality control automation, Defect detection, Kinematic modeling, Dimensional verification, Autonomous manufacturing systems.*

I. INTRODUCTION

Inspection and quality control is an essential component of the modern day manufacturing process, due to an increased need for reliability, safety and performance. Today, manufacturing processes have evolved to become progressively more efficient and product geometries more

complicated, forcing advances in quality control. Traditional inspection methods are unable to maintain pace with the demands of higher output manufacturing practices and tighter tolerances. The increased demand for customized products and rapid production cycles has made traditional quality control methods insufficient, necessitating sophisticated techniques for quality management.[10] This is especially important in industries such as aerospace, automotive, electronics and medical devices, where inspection systems are required to detect defects quickly while minimizing delays and human error.

Early quality control required workers to use manual measurement tools such as calipers and gauges. These methods were sufficient for smaller scaled productions, however is unable to keep up with increased demand of manufacturing nowadays. To keep up with the evolution of manufacturing, two systems were created in the late 1900s. The first being the coordinate measuring system (CMM). The CMM uses a probing system to precisely measure product dimensions. This allowed manufacturers to significantly improve their quality control processes. Machine vision systems further advanced inspection processes by using cameras to automatically inspect products for defects. Together these two were the pioneers of automation in the inspection and quality control departments.

Despite their advantages, these early quality control systems still had their limitations. The CMM required parts to be physically removed from the production line, reducing its efficiency. Early machine vision system were fixed and not very adaptable to a variation of products and complex geometries.

The introduction of robotic inspection systems helped to resolve many of these limitations by combining industrial robots in their production lines. These machines are paired with sensors, vision systems, and other inspection equipment, and are not fixed to a singular spot like the previous systems. These robots can move throughout workspaces inspecting parts from many different angles. They are essentially a moving CMM and machine vision system combined.

Additionally, robotic inspection has begun progressing to not only detect defects but is being integrated with data analytic platforms as well. They can collect real time information from manufacturing processes to help reduce waste, identify defect rates and much more. This data that robotic inspection systems are able to collect are crucial to maintain competitiveness and ensuring an efficient and quality product.

This paper examines the current state of robotic inspect and quality control systems, starting with an overview on the development of inspection methods that led to modern robotic systems. The objective will be to evaluate the strengths and limitations of current approaches, and discuss any developments that may further enhance quality control in modern manufacturing.

II. LITERATURE REVIEW

A. Vision-Based Surface Defect Inspection

Akhyar et al. [1] built the Forceful Defect Detector (FDD) to specifically identify surface defects on steel products. Rather than using an all-purpose object detection network, the authors adapted the backbone to handle the specific visual characteristics of steel flaws by detecting its tunes, which tend to be low-contrast and irregular compared to the typical objects' detection networks are normally trained on. Tested against existing steel defect benchmarks, this model showed better precision detection than earlier model approaches. The bigger takeaway is that defect detection benefits from architectural changes within a domain specific rather than reusing a network built for general object recognition. The paper also discusses deployment limits, since inference speed and false positives can both affect a moving line. This is the type of specialized vision algorithm that is mounted on a robotic arm instead of a fixed camera that becomes the kind of multi angle inspection as described in the introduction.

Zhao et al. [2] built RDD-YOLO on the YOLOv5 architecture for steel surface defect detection. They swapped in a Res2Net backbone for a better multi-scale feature extraction and added a double Feature Pyramid Network to effectively combine features across resolutions. A decoupled detection head splits classification from localization, which the authors say helps with small or subtle defects. On steel defect datasets, the model beat baseline YOLO variants on both accuracy and recall, especially for defects that take up only a micro portion of the image. This fits into a larger pattern of adapting general detection frameworks for industrial inspection, where defect classes often look similar to background texture variation, the same problem that the fixed mount vision systems described in the introduction were never able to fully solve.

Table 1. Performance comparison of RDD-YOLO against baseline YOLOv5 on steel surface defect datasets [2].

Dataset	YOLOv5 mAP	RDD-YOLO mAP	Improvement
NEU-DET	76.8%	81.1%	+4.3%
GC10-DET	69.4%	75.2%	+5.8%

Bagherzadeh et al. [3], covered a broader systematic survey on object-detection-based defect inspection, focused on dataset labeling quality rather than model design. Their analysis found that performance on surface defect tasks depends more on how consistently and precisely training data is labeled than on which architecture is used. They looked at several widely used public datasets, including NEU-DET for steel surfaces and Alibaba's Aluminum Profile Surface Detection Database, which contains thousands of high-resolution images across ten defect categories including dents, pitting, and coating cracks. Their conclusion is that progress in automated inspection depends as much on standardizing datasets as on building new models, which matters directly for any team putting together a custom dataset for a robotic inspection task on a moving line.

A 2025 benchmark study in the Journal of Intelligent Manufacturing [4] does something most defect detection papers did not: instead of proposing a new model, it compares existing ones under controlled, reproducible conditions. The authors pointed out that prior papers often use inconsistent train and test splits, which makes it hard to trust head-to-head comparisons. Their benchmark ran several leading detection architectures, including descendants of RDD-YOLO, through a standardized statistical protocol. The result is that performance gaps between competing models are often smaller, and less statistically meaningful, than the original papers claim once conditions are normalized. For anyone choosing a detection architecture for a robotic inspection task, this is a useful check against the headline numbers from individual papers.

Earlier on a 2023 paper in Robotics and Computer-Integrated Manufacturing [5] presented a multisensory digital twin approach for catching defects during robotic laser-directed energy deposition, a metal additive manufacturing process. Instead of relying on a single camera, the system combines a coaxial melt-pool vision camera with a microphone picking up acoustic signatures of the build, and a short wavelength infrared thermal camera. The main contribution is a method for synchronizing these three sensor streams in time and space with the motion data of the robotic deposition arm, so defects like pores and cracking can be pinned down to a specific location and moment during the build. The conclusion is straightforward: single mode vision systems miss defects that only show up thermally or acoustically, which is the argument for building a multisensory inspection directly into robotic systems.

Later in a 2023 systematic review of deep learning-based surface defect detection [6] covers work published between 2020 and 2023 across various industrial product categories. It sorts approaches into supervised, semi-supervised, and unsupervised categories and notes that supervised object detection still dominates the literature despite needing the largest labeled datasets. One striking figure the authors cited is that inspection eats up over 50 percent of total manufacturing time in some processes, framing automated detection as a throughput problem just as much as a quality one. They also note that some surface defects are invisible to the human eye entirely, which points to a capability that the kind of robotic systems described in the introduction could extend beyond what manual inspection has ever been able to achieve. Dataset imbalance, where defective samples are rare compared to non-defective ones, is flagged as the most persistent open problem in the field.

Table 2. Comparison of defect detection for different approaches per 2023 systematic review [6].

Approach Category	Data Requirement	Pros	Cons
Supervised	Large labeled defect datasets	Highest reported accuracy	Labeling cost high; struggles with rare defect classes
Semi-supervised	Small labeled set +	Reduces labeling burden	Performance sensitive to quality of

	larger unlabeled set	while retaining accuracy	unlabeled data
Unsupervised	No labeled defects required	Useful when defective samples are scarce	Lower precision; higher false-positive rate

B. Modern Inspection and Quality Control Robotic Systems

Modern manufacturing systems are increasingly relying on robotic inspection and quality control systems to meet the increasing need for more efficient production lines and higher quality products. The increased demand for customized products and rapid production cycles has made traditional quality control methods insufficient, necessitating sophisticated techniques for quality management. [9] This has been mostly driven by the introduction of Industry 4.0, also referred to as the fourth industrial revolution. This revolution integrates technologies such as artificial intelligence, robotics, cloud computing, machine vision, and Industrial Internet of Things (IIoT) into manufacturing processes. These systems facilitate smooth automation, data-based decision-making and real-time response in all production processes [8]. Through this integration of interconnected smart tools, these technologies allow machines and sensors to exchange data live. This allows manufacturers to improve production control, maintain higher levels of product consistency, and require less dependency on human operators.

One of the most widely used technologies in modern inspection systems is machine vision, these utilize high resolution cameras, light sensors, and laser sensors to identify dimensions, surface defects, and assembly errors. Modern advances in machine vision utilize AI to analyze patterns from datasets to improve detection. [11] There is a subset of artificial intelligence that utilizes a network with multiple processing layers to learn these complex patterns. This is beneficial compared to traditional machine learning, because it doesn't rely on manually programmed rules. Deep learning models can learn features directly from data without direct human interaction. These networks are exposed to millions of examples of both acceptable and defective products, allowing it to recognize certain characteristics.

Industrial robotic arms have become a great platform for automated tasks. Modern six-axis robots have very high accuracy and repeatability what including cameras, laser scanners, ultrasonic sensors, and even coordinate measuring probes. These robots can execute programmed inspection plans and collect data from multiple angles. These can be utilized in automotive manufacturing to measure weld quality, body. Many other industries also use these robotic arms in many ways. Another form of robotic arm is collaborative robots. Unlike conventional robots that need a safety cage, collaborative robots are designed to work safely alongside humans. [13] They have integrated force sensors and advanced control systems that make it safe to share workspaces, so that humans can perform tasks while robots handle repetitive inspection procedures.

Additionally, autonomous mobile robots have expanded the inspection and quality control market beyond fixed stations. These are equipped with navigation systems, cameras

and LiDAR sensors. These machines can travel throughout production facilities to perform inspections of equipment, production, and inventory. These systems support preventative maintenance by monitoring machine conditions and identifying potential failures before they lead to downtime. These robots are being utilized in hazardous environments such as nuclear facilities and chemical processing plants where it may be dangerous for humans.

Many of these robotic systems is the use of multiple sensors at once. Rather than relying on a single sensing system many advanced robotic inspection platforms combine information from cameras, LiDAR sensors, thermal systems, and ultrasonic sensors. By using data from multiple sources, these systems can provide a much more comprehensive understanding of a products condition. [12] Visual sensors identify surface defects such as scratches, while ultrasonic sensors can detect flaws such as cracks that are not visible to the human eye. This can significantly improve quality control within these robotic systems. Cloud computing has further enhanced the capabilities of robotic inspection systems. Data can now be transmitted to cloud platforms for easier access. This enables manufacturers to monitor quality across many facilities in real time and identify quality issues.

As these systems become more autonomous, researches are also studying adaptive robotic control which would allow for robots to modify their own inspection programs, and data collection based on their observations. This greatly improves flexibility in these robots and reduces the need for human interaction. These developments should play a huge role in the next generation of quality control, where robots can not only identify defects, but adapt to changing requirements.

Despite many advancements, modern robotic inspection systems still face a few challenges. Deep learning systems require huge quantities of described data for training, because they need different lighting conditions and product variations. There is a high implementation cost and complexity that may limit it among smaller manufacturers. Current researching is focusing on reducing training requirements, enhancing sensor fusion, and developing more flexible autonomous inspection processes. These advancements will further expand inspection and quality control systems.

C. Force-Controlled and Collaborative Robots for Contact-Based Inspection

A 2024 paper presents ACES, a teleoperated robotic system built for inspecting the inside of pipes, which uses a force-controlled six-degree-of-freedom robotic arm as part of its measurement setup [15]. The arm carries a contact sensor that can detect and locate contact points at the end effector relative to the robot's base frame. To measure the inner geometry of a pipe section, the manipulator touches five points on the cylindrical wall, as long as no four of those points lie in the same plane, and from these five contact coordinates the system can fully determine the inner cylinder's dimensions and orientation within the robot's base frame. A measurement tool then traces a circular path centered on the cylinder axis to complete the survey. The authors note that despite the generality of this approach, it comes with real drawbacks: the system is expensive, and the control involved is complex, especially when the pipe geometry includes elbows or branching sections.

A 2025 study in *Cyborg and Bionic Systems* presents a robotic ultrasound scanning system with an adjustable constant-force end-effector [16]. The core problem the authors address is that in manual ultrasound scanning, the contact force between the probe and the patient's skin depends entirely on the operator's skill and experience, which creates inconsistency in image quality. Their solution combines a passive mechanism, which absorbs sudden force variations like a spring or cushion would, with an active force control system that handles long-range positioning and fine adjustments. Tests on both silicone models and human volunteers showed that the system could produce high-quality ultrasound images with strong repeatability. While this work is framed around medical imaging rather than industrial inspection, the underlying problem, maintaining a consistent contact force between a sensor and a surface that is not perfectly flat or rigid, is identical to what a robotic inspection arm faces when running an ultrasonic thickness probe across an industrial part.

A 2023 paper on co-robotic ultrasound for mammography breaks the robot's motion into three distinct phases, which is a useful framework for thinking about any contact-based inspection task [17]. First, the robot positions the probe above the target location using a calculated offset along the surface normal. Second, the probe moves downward until it makes contact, monitored by a six-axis force and torque sensor mounted between the end effector and the probe, stopping once contact force reaches a set threshold. Third, the robot performs a linear scan while a proportional-integral-derivative controller keeps the contact force within bounds throughout the motion. The scanning velocity is split into a component moving along the surface and a component moving into or away from the surface, allowing the robot to track gentle changes in surface shape without losing contact or pressing too hard. This three-phase structure, approach, contact, scan, maps directly onto how a robotic probe would inspect a curved or uneven industrial component.

A 2024 paper on automatic contact-based 3D scanning describes an open-loop, six-degree-of-freedom articulated robotic system that uses a digitizer probe to detect contact with an object's surface [18]. Inverse kinematics is used to figure out what joint angles position the probe correctly, and straight-line trajectory planning controls the motion between points. The system can take single-point measurements or scan entire freeform surfaces, and once a user specifies the size, position, and density of the area to be scanned, the system handles the rest automatically. Results are output as 3D scans in a standard format compatible with common 3D modeling software. The authors tested the system's accuracy and repeatability against an established measurement standard and compared the resulting point cloud to the object's original 3D model to check how closely the scan matched reality. This paper is a clear, practical example of how the kinematic concepts from Craig's textbook, particularly inverse kinematics and trajectory planning, are applied directly to build a working contact-based inspection tool.

A 2025 review in the *Journal of Intelligent Manufacturing* looks at how collaborative robots, or cobots, are being used specifically for quality control tasks [19]. The authors note that cobots have mostly been used for material handling, assembly, and positioning, while their role in inspection has been comparatively underexplored until recently. One finding stands out: between 2014 and 2023, the balance between

offline inspection (checking parts after they're removed from the line) and in-process inspection (checking parts while production continues) shifted from roughly even to a 90/10 split favoring in-process methods. The review also describes a real manufacturing example where a company used camera-equipped cobots to inspect parts directly on a lathe, which let them increase their workforce by 50 percent and grow sales by 70 percent, since the cobots took over inspection tasks that previously consumed operator time. The authors frame this shift toward in-process inspection as part of a broader move toward zero-defect manufacturing, where problems are caught and corrected before they ever leave the workstation.

A 2024 paper in the *Journal of Robotics* examines the broader role of cobots within what's being called Industry 5.0, the next stage of manufacturing that puts renewed emphasis on human-robot collaboration rather than full automation [20]. The paper surveys how cobots are being integrated across industrial processes and discusses the technical, economic, and social challenges that come with that integration. One point relevant to inspection specifically is the emphasis on regular maintenance and inspection of the cobots themselves, to make sure their safety systems, particularly the force and torque sensors that allow them to work near humans without cages, continue functioning correctly. This creates an interesting loop: cobots are increasingly used to inspect other equipment, but they also require their own inspection regime to remain safe collaborators. The paper's broader conclusion is that successful cobot integration depends as much on addressing workforce trust and training as on the underlying technology itself.

III. PREPARE YOUR PAPER BEFORE STYLING

- [1] F. Akhyar, Y. Liu, C.-Y. Hsu, T. K. Shih, and C.-Y. Lin, "FDD: a deep learning-based steel defect detectors," *Int. J. Adv. Manuf. Technol.*, vol. 126, no. 3, pp. 1093-1107, 2023.
- [2] C. Zhao, X. Shu, X. Yan, X. Zuo, and F. Zhu, "RDD-YOLO: A modified YOLO for detection of steel surface defects," *Measurement*, vol. 214, p. 112776, 2023.
- [3] F. Bagherzadeh, T. Shafighfar, R. M. A. Khan, P. Szczuko, and M. Mieloszyk, et al., "A systematic review of deep learning approaches for surface defect detection in industrial applications," *Eng. Appl. Artif. Intell.*, 2023.
- [4] Y. Ma et al., "Surface defect inspection of industrial products with object detection deep networks: a systematic review," *Artif. Intell. Rev.*, 2024.
- [5] L. Chen, G. Bi, X. Yao, C. Tan, J. Su, N. P. H. Ng, Y. Chew, K. Liu, and S. K. Moon, "Multisensor fusion-based digital twin for localized quality prediction in robotic laser-directed energy deposition," *Robot. Comput.-Integr. Manuf.*, vol. 84, p. 102581, 2023.
- [6] Y. Ma et al., "Surface defect inspection of industrial products with object detection deep networks: a systematic review," *Artif. Intell. Rev.*, 2024.
- [7] K. G. Amin, M. Patidar, P. K. Patidar, N. K. R. Rafaliya, J. Joshi and D. Mishra, "A Review on Robotics and Automation Techniques for Quality Inspection in Industry 4.0," 2025 IEEE International Conference on Advances in Computing Research On Science Engineering and Technology (ACROSET), INDORE, India, 2025, pp. 1-5, doi: 10.1109/ACROSET66531.2025.11280985.
- [8] A. Villalonga, Y. J. Cruz, D. Alfaro, R. E. Haber, J. L. Martínez-Lastra and F. Castaño, "Enhancing Quality Inspection in Zero-Defect Manufacturing Through Robotic-Machine Collaboration," 2024 7th Iberian Robotics Conference (ROBOT), Madrid, Spain, 2024, pp. 1-6, doi: 10.1109/ROBOT61475.2024.10797427
- [9] A. E. Duron, B. A. C. Stiles and D. A. Torck, "RIPTIDE: Robot Inspecting Parts to Increase Development Efficiency," 2025 22nd International Conference on Ubiquitous Robots (UR), College Station, TX, USA, 2025, pp. 333-339, doi: 10.1109/UR65550.2025.11078063.

- [10] Brito, T., Queiroz, J., Piardi, L., Fernandes, L. A., Lima, J., & Leitão, P. (2020). A Machine Learning Approach for Collaborative Robot Smart Manufacturing Inspection for Quality Control Systems. *Procedia Manufacturing*, 51, 11–18. <https://doi.org.ezproxywit.flo.org/10.1016/j.promfg.2020.10.003>
- [11] Zhou, M., Ge, C., & Xiong, X. (2025). Research on a Rail-Mounted Concrete Quality Inspection Robot System for Aerial Building Machine. *2025 10th International Conference on Automation, Control and Robotics Engineering (CACRE), Automation, Control and Robotics Engineering (CACRE), 2025 10th International Conference On*, 9–16. <https://doi.org.ezproxywit.flo.org/10.1109/CACRE66141.2025.11119617>
- [12] Discepolo, S., & Castellini, P. (2024). Neural Network to Retrieve Features of a Mechanical Component Acquired by a Robotic Inspection System. *2024 IEEE International Conference on Metrology for EXtended Reality, Artificial Intelligence and Neural Engineering (MetroXRAINE), Metrology for EXtended Reality, Artificial Intelligence and Neural Engineering (MetroXRAINE), 2024 IEEE International Conference On*, 325–330. <https://doi.org.ezproxywit.flo.org/10.1109/MetroXRAINE62247.2024.10797005>
- [13] Czimmermann, T., Chiurazzi, M., Milazzo, M., Roccella, S., Barbieri, M., Dario, P., Oddo, C. M., & Ciuti, G. (2022). An Autonomous Robotic Platform for Manipulation and Inspection of Metallic Surfaces in Industry 4.0. *IEEE Transactions on Automation Science and Engineering, Automation Science and Engineering, IEEE Transactions on, IEEE Trans. Automat. Sci. Eng.*, 19(3), 1691–1706. <https://doi.org.ezproxywit.flo.org/10.1109/TASE.2021.3122820>
- [14] Irvine, D., & Narayanan, M. (1995). Inspection and quality control-an overview. *IEEE Technical Applications Conference and Workshops. Northcon/95. Conference Record, Northcon 95. I EEE Technical Applications Conference and Workshops Northcon95, Northcon/95*, 376–380. <https://doi.org.ezproxywit.flo.org/10.1109/NORTHCON.1995.485100>
- [15] "ACES: A teleoperated robotic solution to pipe inspection from the inside," arXiv preprint arXiv:2405.09925, 2024.
- [16] Z. Wu et al., "Robotic ultrasound scanning end-effector with adjustable constant contact force," *Cyborg Bionic Syst.*, 2025.
- [17] "Enabling mammography with co-robotic ultrasound," arXiv preprint arXiv:2312.10309, 2023.
- [18] "Automatic contact-based 3D scanning using articulated robotic arm," presented at Int. Conf. on Interactive Design and Manufacturing, 2024.
- [19] "Collaborative robots for quality control: an overview of recent studies and emerging trends," *J. Intell. Manuf.*, 2025.
- [20] M. Al-Amin, "Cobotics: The evolving roles and prospects of next-generation collaborative robots in Industry 5.0," *J. Robot.*, vol. 2024, Art. no. 2918089, 2024.